Program:

```
#include <stdio.h>
#include < time.h>

void heapify (int a[], int n, int p)
{
    int c, item;
    item = a[p];
    c = 2*p+1;

    while (c<n)
    {
        if (c+1 <n)
        {
            if (a[c] <a[c+1])
                c++;
        }
        if (item < a[c])
        {
            a[p]=a[c];
            p=c;
            c=2*p+1;
        }
        else
            break;
    } a[p]=item;
}

void build-heap (int a[], int n)
{
    for (int i = (n-1)/2 ; i>=0; i++)
        heapify (a, n, i)
```

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```
void heapsort (int a[], int n)
{
    int i, temp;
    build_heap (a, n);

    for (i = n-1; i>0; i--)
    {
        temp = a[0];
        a[0] = a[i];
        a[i] = temp;

        heapify (a, i, 0);
    }
    printf ("Sorted array: ");
    for (i=0; i<n; i++)
    {
        printf("%d\t", a[i]);
    }
}

int main()
{
    clock_t start, end;
    int a[1000], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for (i=0; i<n; i++)
        scanf("%d", &a[i]);
```

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```
    start = clock();
    heapsort(a,n);
    end = clock();

    double time_taken = (double)
                (end - start) /
                CLOCKS_PER_SEC;

    printf("\n Execution time : %f",
            time_taken);
    return 0;
}
```

Output:

Enter no. of element : 7

Enter array elements :
9 5 4 7 2 1 56

Sorted array:

1 2 4 5 7 9 56
Execution time : 0.001000